🚀 MY FUTURE IN CODING: CAREER PATHS, JOBS & OPPORTUNITIES 💻🐍

Name: Christabel
Current School Level: SS2, First Term
Main Programming Language: Python
Other Skills: HTML and CSS
Currently Exploring: Game Development
Additional Skill I Plan to Learn: CorelDRAW
Future Goal: Become a skilled programmer and Computer Science student, build useful projects, participate in competitions, earn recognition, and explore a successful career in technology.

🌟 INTRODUCTION

Learning to code can lead to many different careers.

I do not have to become only a "Python programmer." Python is a foundation that can lead to careers in:

- Software development
- Web development
- Mobile app development
- Game development
- Artificial intelligence
- Machine learning
- Data analysis
- Data science
- Cybersecurity
- Cloud computing
- DevOps
- Database engineering
- Robotics
- Embedded systems
- Computer research
- Creative technology
- Technical art
- Technology entrepreneurship

I also do not have to choose my final career immediately.

While I am still in secondary school, I can explore different areas and discover which ones I enjoy most.

💻 MY POSSIBLE CAREER PATH

My current learning journey can eventually look like this:

🐍 **Python**
↓
🧠 **Computer Science Fundamentals**
↓
📚 **Algorithms & Data Structures**
↓
🌐 **Revise HTML & CSS**
↓
⚡ **Learn JavaScript**
↓
💻 **Build Real Projects**
↓
🏆 **Enter Competitions**
↓
🎮 **Explore Game Development**
↓
🤖 **Explore AI & Data**
↓
🎨 **Learn CorelDRAW as a Creative Side Skill**
↓
🚀 **Choose a Specialization**

💰 IMPORTANT NOTE ABOUT INCOME

The income figures in this note are **rough career-planning estimates**, not guaranteed salaries.

For consistency, the approximate conversion used in this note is:

$1 ≈ ₦1,365

The naira figures are simply approximate conversions of the dollar figures.

Actual income depends on:

- Country
- Company
- Experience
- Skill level
- Education

- Specialization
- Job responsibilities
- Remote or local employment
- Freelancing or full-time employment
- Ability to solve real-world problems

A person working for a Nigerian company may earn much less than the converted U.S. salary.

A person working remotely for an international company may earn closer to international salary levels.

Exchange rates also change over time.

💻 1. SOFTWARE DEVELOPER / SOFTWARE ENGINEER

What They Do

Software developers and software engineers build applications, software systems, tools, and computer programs.

They may:

- Create new software
- Build applications
- Fix bugs
- Improve existing software
- Design software features
- Work with databases
- Work with APIs
- Collaborate with other programmers and engineers

Common Programming Languages and Tools

- Python
- JavaScript
- TypeScript
- Java
- C++
- C#
- SQL
- Git
- GitHub

Example Companies

- Microsoft
- Google
- Amazon
- Apple
- Meta
- Banks
- Fintech companies
- Startups
- Software companies

Approximate Income

Intern: $500–$2,000/month
≈ ₦682,500–₦2.73 million/month

Junior/Beginner: $1,000–$4,000/month
≈ ₦1.365 million–₦5.46 million/month

Mid-level: $4,000–$8,000/month
≈ ₦5.46 million–₦10.92 million/month

Senior/Expert: $8,000–$15,000+/month
≈ ₦10.92 million–₦20.48 million+/month

🌐 2. WEB DEVELOPER

What They Do

Web developers create websites and web applications.

They can specialize in:

- Front-end development
- Back-end development
- Full-stack development

Common Programming Languages and Tools

- HTML
- CSS
- JavaScript
- TypeScript

- React
- Node.js
- Python
- Django
- Flask
- SQL

Example Companies

- Google
- Microsoft
- Amazon
- Shopify
- Netflix
- Banks
- Fintech companies
- Startups
- Schools
- Organizations with online platforms

Approximate Income

Intern: $300–$1,000/month
≈ ₦409,500–₦1.365 million/month

Junior/Beginner: $700–$2,500/month
≈ ₦955,500–₦3.41 million/month

Mid-level: $2,500–$6,000/month
≈ ₦3.41 million–₦8.19 million/month

Senior: $6,000–$12,000+/month
≈ ₦8.19 million–₦16.38 million+/month

My Connection to This Career

I already know HTML and CSS, so web development is a natural area for me to explore after learning JavaScript.

📱 3. MOBILE APP DEVELOPER

What They Do

Mobile app developers create applications for smartphones and tablets.

They may build:

- Android apps
- iPhone apps
- Business apps
- Social media apps
- Banking apps
- Educational apps
- Games

Common Programming Languages and Tools

- Kotlin
- Java
- Swift
- Dart
- Flutter
- React Native

Example Companies

- Apple
- Google
- Samsung Electronics
- Meta
- Spotify
- Banks
- Fintech companies
- Startups

Approximate Income

Intern: $500–$1,500/month
≈ ₦682,500–₦2.05 million/month

Junior/Beginner: $1,000–$3,500/month
≈ ₦1.365 million–₦4.78 million/month

Mid-level: $3,500–$7,000/month
≈ ₦4.78 million–₦9.56 million/month

Senior: $7,000–$12,000+/month
≈ ₦9.56 million–₦16.38 million+/month

🎮 4. GAME DEVELOPER

What They Do

Game developers create video games.

They may work on:

- Gameplay
- Game mechanics
- Player controls
- Physics
- Game systems
- Graphics integration
- Sound integration
- Artificial intelligence for games

Common Programming Languages and Tools

- C#
- C++
- Python
- Unity
- Unreal Engine
- Godot

Example Companies

- Epic Games
- Unity Technologies
- Electronic Arts
- Nintendo
- Sony Interactive Entertainment
- Ubisoft

Approximate Income

Intern: $300–$1,000/month
≈ ₦409,500–₦1.365 million/month

Junior/Beginner: $700–$2,500/month
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≈ ₦8.19 million–₦16.38 million+/month

My Connection to This Career

I am currently exploring game development, so I want to continue learning about it and eventually create my own original game.

🤖 5. AI / MACHINE LEARNING ENGINEER

What They Do

AI and machine learning engineers build systems that can learn from data.

They may work on:

- Artificial intelligence
- Recommendation systems
- Computer vision
- Natural-language systems
- Predictive models
- Machine learning systems

Common Programming Languages and Tools

- Python
- PyTorch
- TensorFlow
- scikit-learn
- NumPy
- pandas
- SQL

Example Companies

- OpenAI
- Google DeepMind
- NVIDIA
- Microsoft
- Amazon
- Meta

Approximate Income

Intern: $500–$2,000/month
≈ ₦682,500–₦2.73 million/month

Junior/Beginner: $1,500–$4,000/month
≈ ₦2.05 million–₦5.46 million/month

Mid-level: $4,000–$10,000/month
≈ ₦5.46 million–₦13.65 million/month

Senior/Expert: $10,000–$20,000+/month
≈ ₦13.65 million–₦27.3 million+/month

Important Skills

To eventually work in this field, I would need:

- Strong Python
- Mathematics
- Statistics
- Algorithms
- Data structures
- Machine learning concepts

📊 6. DATA ANALYST / DATA SCIENTIST

What They Do

Data professionals work with information and data.

They may:

- Analyze data
- Find patterns
- Create reports
- Visualize data
- Help companies make decisions
- Build predictive models

Common Programming Languages and Tools

- Python

- SQL
- Excel
- Power BI
- Tableau
- R
- pandas
- NumPy

Example Companies

- Google
- Microsoft
- Amazon
- JPMorgan Chase
- Deloitte
- Banks
- Fintech companies
- Healthcare organizations
- Consulting firms
- Research institutions

Approximate Income

Intern: $300–$1,000/month
≈ ₦409,500–₦1.365 million/month

Junior/Beginner: $700–$2,500/month
≈ ₦955,500–₦3.41 million/month

Mid-level: $2,500–$7,000/month
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Senior: $7,000–$15,000+/month
≈ ₦9.56 million–₦20.48 million+/month

🔐 7. CYBERSECURITY SPECIALIST

What They Do

Cybersecurity specialists help protect computer systems, networks, and data.

They may:

- Monitor systems

• Find security weaknesses
• Investigate security incidents
• Protect networks
• Improve security systems
• Help organizations respond to cyber threats

Common Programming Languages and Tools

• Python
• Linux
• Bash
• PowerShell
• SQL
• Networking tools
• Cybersecurity tools

Example Companies

• Palo Alto Networks
• CrowdStrike
• Cisco
• Microsoft
• Banks
• Fintech companies
• Government organizations
• Cybersecurity firms

Approximate Income

Intern: $300–$1,000/month
≈ ₦409,500–₦1.365 million/month

Junior/Beginner: $1,000–$3,500/month
≈ ₦1.365 million–₦4.78 million/month

Mid-level: $3,500–$8,000/month
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Senior/Expert: $8,000–$15,000+/month
≈ ₦10.92 million–₦20.48 million+/month

☁️ 8. CLOUD / DEVOPS ENGINEER

What They Do

Cloud and DevOps engineers help companies deploy, run, and maintain software and infrastructure.

They work with:

- Servers
- Cloud computing
- Software deployment
- Automation
- Infrastructure
- Continuous integration and delivery

Common Programming Languages and Tools

- Python
- Linux
- Docker
- Kubernetes
- AWS
- Microsoft Azure
- Google Cloud
- GitHub Actions

Example Companies

- Amazon Web Services
- Microsoft
- Google
- IBM
- Banks
- Fintech companies
- Large technology companies

Approximate Income

Intern: $500–$1,500/month
≈ ₦682,500–₦2.05 million/month

Junior/Beginner: $1,000–$4,000/month
≈ ₦1.365 million–₦5.46 million/month

Mid-level: $4,000–$9,000/month
≈ ₦5.46 million–₦12.29 million/month

Senior/Expert: $9,000–$17,000+/month
≈ ₦12.29 million–₦23.21 million+/month

🗄️ 9. DATABASE ENGINEER

What They Do

Database engineers design and manage systems that store and organize important information.

They work with:

- Databases
- Data storage
- Data organization
- Database performance
- Data security

Common Programming Languages and Tools

- SQL
- PostgreSQL
- MySQL
- MongoDB
- Oracle
- Python

Example Companies

- Oracle
- Amazon
- Microsoft
- Google
- Banks
- Fintech companies
- Healthcare organizations
- Large enterprises

Approximate Income

Intern: $400–$1,200/month
≈ ₦546,000–₦1.64 million/month

Junior/Beginner: $1,000–$3,500/month
≈ ₦1.365 million–₦4.78 million/month

Mid-level: $3,500–$8,000/month
≈ ₦4.78 million–₦10.92 million/month

Senior: $8,000–$15,000+/month
≈ ₦10.92 million–₦20.48 million+/month

🦾 10. ROBOTICS / EMBEDDED SYSTEMS ENGINEER

What They Do

Robotics and embedded systems engineers program robots and electronic devices.

They may work with:

- Robots
- Sensors
- Microcontrollers
- Electronic devices
- Automated systems

Common Programming Languages and Tools

- C
- C++
- Python
- Arduino
- Raspberry Pi
- ROS
- Microcontrollers

Example Companies

- NVIDIA
- Tesla
- Boston Dynamics
- iRobot
- Aerospace companies
- Automobile companies
- Robotics companies
- Research institutions

Approximate Income

Intern: $300–$1,000/month
≈ ₦409,500–₦1.365 million/month

Junior/Beginner: $700–$3,000/month
≈ ₦955,500–₦4.095 million/month

Mid-level: $3,000–$7,000/month
≈ ₦4.095 million–₦9.56 million/month

Senior: $7,000–$15,000+/month
≈ ₦9.56 million–₦20.48 million+/month

🧪 11. COMPUTER RESEARCH SCIENTIST

What They Do

Computer research scientists work on advanced computing research.

They may research:

- New algorithms
- Artificial intelligence
- Machine learning
- Computing methods
- New technologies

Common Programming Languages and Tools

- Python
- C++
- Java
- MATLAB
- Research frameworks

Example Companies and Organizations

- Google DeepMind
- OpenAI
- Microsoft Research
- NVIDIA
- Universities
- Research laboratories

Approximate Income

Research Intern: $500–$2,000/month
≈ ₦682,500–₦2.73 million/month

Entry-level: $1,500–$4,000/month
≈ ₦2.05 million–₦5.46 million/month

Experienced: $4,000–$10,000/month
≈ ₦5.46 million–₦13.65 million/month

Senior Researcher: $10,000–$20,000+/month
≈ ₦13.65 million–₦27.3 million+/month

Important

This career often requires advanced university education and research experience.

🎨 12. CREATIVE TECHNOLOGIST / TECHNICAL ARTIST

What They Do

Creative technologists and technical artists combine programming, art, design, animation, and technology.

They may work on:

- Games
- Animation
- Visual effects
- Interactive experiences
- Digital art
- Technical tools for artists

Common Programming Languages and Tools

- Python
- C#
- C++
- Unity
- Unreal Engine
- Blender

- CorelDRAW
- Photoshop

Example Companies

- Epic Games
- Electronic Arts
- Ubisoft
- Adobe
- Animation studios
- Advertising agencies
- Game studios
- Media companies

Approximate Income

Intern: $300–$1,000/month
≈ ₦409,500–₦1.365 million/month

Junior/Beginner: $700–$2,500/month
≈ ₦955,500–₦3.41 million/month

Mid-level: $2,500–$6,000/month
≈ ₦3.41 million–₦8.19 million/month

Senior: $6,000–$12,000+/month
≈ ₦8.19 million–₦16.38 million+/month

My Connection to This Career

This could be an interesting combination for me because I am exploring:

Programming + Game Development + CorelDRAW + Creativity

🚀 13. TECH ENTREPRENEUR / STARTUP FOUNDER

What They Do

A technology entrepreneur creates a technology product, service, or company.

They might create:

- Software applications

- Websites
- Mobile apps
- AI products
- Games
- Online platforms
- Technology companies

Common Programming Languages and Tools

There is no single required language.

The technology depends on the product.

Possible technologies include:

- Python
- JavaScript
- Java
- C#
- C++
- Databases
- AI tools
- Mobile development tools

Companies

The goal is eventually to create **my own company**.

Income

There is no fixed salary.

Beginning: Could be **$0 / ₦0**

Small successful business: Could earn hundreds or thousands of dollars per month.

Highly successful startup: Could generate millions of dollars in revenue.

However, entrepreneurship is unpredictable. A business can also fail or lose money.

🏆 MY CAREER EXPLORATION MAP

Based on my current interests, these are the areas I want to explore.

🥇 SOFTWARE DEVELOPMENT

Python → JavaScript → Git/GitHub → Projects

🥈 ARTIFICIAL INTELLIGENCE / MACHINE LEARNING

Python → Mathematics → Statistics → Data Structures → Machine Learning

🥉 WEB DEVELOPMENT

HTML + CSS Revision → JavaScript → React → Backend Development

🎮 GAME DEVELOPMENT

Programming Fundamentals → C# → Game Engine → Build My Own Game

🏆 COMPETITIVE PROGRAMMING

Python → Algorithms → Data Structures → Problem-Solving → Competitions

🎨 CREATIVE TECHNOLOGY

Programming + CorelDRAW + Design + Game Development

🌟 MY PERSONAL LONG-TERM GOAL

I want to become an excellent programmer and study Computer Science at university.

I want to:

☐ Learn Python deeply
☐ Learn JavaScript
☐ Revise HTML and CSS
☐ Explore game development
☐ Learn CorelDRAW with my mom
☐ Build useful projects
☐ Participate in programming competitions
☐ Participate in technology competitions
☐ Earn medals and recognition
☐ Build a strong portfolio
☐ Make my parents proud

☐ Explore different technology careers
☐ Eventually choose a specialization that I truly enjoy

I do not need to choose my final career immediately.

My goal right now is to become **really good at learning, problem-solving, programming, and building things**.

The skills I build now will give me more choices in the future.

💡 MY BIGGEST LESSON

I do not have to know exactly what I will become today.

I can explore.

I can learn.

I can make mistakes.

I can build projects.

I can enter competitions.

I can discover what I enjoy.

And as I grow, I can choose the path that fits me best.

🐍 **Learn.** 💻 **Build.** 🧠 **Solve.** 🏆 **Compete.** 🎨 **Create.** 🚀 **Grow.**

This is the beginning of my journey into the world of technology.